

According to the New Book & Syllabus of
Board of Intermediate, Sindh.

THE STUDENTS' COMPANION OF *Mathematics*

For Class **11**

INCLUDES **MODEL PAPER** FOR FURTHER PRACTICE



Chapter # 05 MISCELLANEOUS SERIES

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CHAPTER # 05 MISCELLANEOUS SERIES**EXERCISE 5.1**

Sum the following series up to indicated terms.

Q.1 $1^2 + 2^2 + 3^2 + \dots + 10^2$

SOLUTION:

$$\sum n^2 = \frac{n(n+1)(2n+1)}{6}$$

$$\sum 10^2 = \frac{10(10+1)(2 \times 10 + 1)}{6} = 385$$

Q.2 $1^3 + 2^3 + 3^3 + \dots + 15^3$

SOLUTION:

$$\sum n^3 = \left[\frac{n(n+1)}{2} \right]^2$$

$$\sum 15^3 = \left[\frac{15(15+1)}{2} \right]^2 = 14400$$

Q.3 $1^2 + 3^2 + 5^2 + \dots$ up to n terms

SOLUTION:Here n th term is $a_n = (2n-1)^2 \Rightarrow a_n = 4n^2 - 4n + 1$

$$\sum (4n^2 - 4n + 1) = 4 \sum n^2 - 4 \sum n + \sum n^0$$

$$= 4 \left(\frac{n(n+1)(2n+1)}{6} \right) - 4 \left(\frac{n(n+1)}{2} \right) + n$$

$$= \frac{2n(n+1)(2n+1)}{3} - 2n(n+1) + n$$

$$= n \left\{ \frac{2(2n^2 + 3n + 1) - 6(n+1) + 3}{3} \right\}$$

$$= n \left\{ \frac{4n^2 + 6n + 2 - 6n - 6 + 3}{3} \right\}$$

$$= n \left\{ \frac{4n^2 - 1}{3} \right\}$$

$$= n \left\{ \frac{(2n)^2 - (1)^2}{3} \right\}$$

$$= n \left\{ \frac{(2n + 1)(2n - 1)}{3} \right\}$$

Q.4 $2^2 + 4^2 + 6^2 + 8^2 + \dots$ up to n terms

SOLUTION:

Here n th term is $a_n = (2n)^2 \Rightarrow a_n = 4n^2$

$$\sum (4n^2) = 4 \sum n^2$$

$$= 4 \left(\frac{n(n+1)(2n+1)}{6} \right)$$

$$= \frac{2n(n+1)(2n+1)}{3}$$

Q.5 $1^3 + 3^3 + 5^3 + \dots$ up to n terms

SOLUTION:

Here n th term is $a_n = (2n - 1)^3 \Rightarrow a_n = 8n^3 - 12n^2 + 6n - 1$

$$\sum (8n^3 - 12n^2 + 6n - 1)$$

$$= 8 \sum n^3 - 12 \sum n^2 + 6 \sum n - \sum n^0$$

$$= 8 \left(\frac{n(n+1)}{2} \right)^2 - 12 \left(\frac{n(n+1)(2n+1)}{6} \right) + 6 \left(\frac{n(n+1)}{2} \right) - n$$

$$= \frac{8n^2(n+1)^2}{4} - 2n(n+1)(2n+1) + 3n(n+1) - n$$

$$= n \{ 2n(n+1)^2 - 2(n+1)(2n+1) + 3(n+1) - 1 \}$$

$$= n \{ 2n(n^2 + 2n + 1) - 2(2n^2 + 3n + 1) + 3n + 3 - 1 \}$$

$$= n \{ 2n^3 + 4n^2 + 2n - 4n^2 - 6n - 2 + 3n + 2 \}$$

$$= n(2n^3 - n)$$

$$= n^2(2n^2 - 1)$$

Q.6 $2 \times 1^2 + 5 \times 2^2 + 8 \times 3^2 + \dots$ up to n terms

SOLUTION:

$$2, 5, 8, \dots$$

$$n^{\text{th}} \text{ term} = 2 + 3(n - 1) = 2 + 3n - 3 = 3n - 1$$

$$\text{Here } n^{\text{th}} \text{ term is } a_n = n^2(3n - 1) \Rightarrow a_n = 3n^3 - n^2$$

$$\sum (3n^3 - n^2) = 3 \sum n^3 - \sum n^2$$

$$= 3 \left(\frac{n(n+1)}{2} \right)^2 - \frac{n(n+1)(2n+1)}{6}$$

$$= \frac{3n^2(n+1)^2}{4} - \frac{n(n+1)(2n+1)}{6}$$

$$= n(n+1) \left\{ \frac{3n(n+1)}{4} - \frac{(2n+1)}{6} \right\}$$

$$= n(n+1) \left\{ \frac{9n(n+1) - 2(2n+1)}{12} \right\}$$

$$= n(n+1) \left\{ \frac{9n^2 + 9n - 4n - 2}{12} \right\}$$

$$= n(n+1) \left\{ \frac{9n^2 + 5n - 2}{12} \right\}$$

Q.7 $1 + (1 + 2) + (1 + 2 + 3) + \dots$ up to n terms

SOLUTION:

$$1 + (1 + 2) + (1 + 2 + 3) + \dots \text{ up to } n \text{ terms}$$

$$= \sum 1 + \sum 2 + \sum 3 + \dots \text{ up to } n \text{ terms}$$

$$a_n = \sum n = \frac{n(n+1)}{2} = \frac{n^2 + n}{2}$$

$$\begin{aligned} \sum \left(\frac{n^2 + n}{2} \right) &= \frac{1}{2} \left\{ \sum n^2 + \sum n \right\} \\ &= \frac{1}{2} \left\{ \frac{n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right\} \\ &= \frac{n(n+1)}{2 \times 2} \left\{ \frac{2n+1}{3} + 1 \right\} \\ &= \frac{n(n+1)}{4} \left\{ \frac{2n+1+3}{3} \right\} \\ &= \frac{n(n+1)}{4} \left\{ \frac{2n+4}{3} \right\} \\ &= \frac{n(n+1)}{12} \{2(n+2)\} \\ &= \frac{n(n+1)(n+2)}{6} \end{aligned}$$

Q.8 $2 + (2 + 5) + (2 + 5 + 8) + \dots$ up to n terms

SOLUTION:

$2 + (2 + 5) + (2 + 5 + 8) + \dots$ up to n terms

$2 + 5 + 8 + \dots$ to n terms

$$a = 2, d = 3$$

$$S_n = \frac{n}{2} \{2a + (n-1)d\}$$

$$a_n = \frac{n}{2} \{2(2) + 3(n-1)\} = \frac{n}{2} \{4 + 3n - 3\} = \frac{n}{2} (3n + 1)$$

$$a_n = \frac{1}{2} (3n^2 + n)$$

$$\text{Sum} = \frac{1}{2} \sum (3n^2 + n)$$

$$= \frac{1}{2} \left\{ 3 \sum n^2 + \sum n \right\}$$

$$= \frac{1}{2} \left\{ \frac{3n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} \right\}$$

$$= \frac{1}{2} \left\{ \frac{n(n+1)(2n+1)}{2} + \frac{n(n+1)}{2} \right\}$$

$$= \frac{n(n+1)}{2 \times 2} \{(2n+1) + 1\}$$

$$= \frac{n(n+1)}{4} \{2n+2\}$$

$$= \frac{n(n+1)}{2} \{n+1\}$$

$$= \frac{1}{2} n(n+1)^2$$

Q.9(i) Find the sum of n terms of the series whose n th term is:

$$3n^2 + n + 1$$

SOLUTION:

$$\text{Sum} = \sum (3n^2 + n + 1)$$

$$= 3 \sum n^2 + \sum n + \sum n^0$$

$$= \frac{3n(n+1)(2n+1)}{6} + \frac{n(n+1)}{2} + n$$

$$= \frac{n}{2} \{(n+1)(2n+1) + (n+1) + 2\}$$

$$= \frac{n}{2} \{(2n^2 + 3n + 1) + n + 3\}$$

$$= \frac{1}{2} n(2n^2 + 4n + 4)$$

$$= n(n^2 + 2n + 2)$$

Q.9(ii) Find the sum of n terms of the series whose n th term is:

$$n^2 + 4n + 1$$

SOLUTION:

$$\begin{aligned} \text{Sum} &= \sum (n^2 + 4n + 1) \\ &= \sum n^2 + 4 \sum n + \sum n^0 \\ &= \frac{n(n+1)(2n+1)}{6} + \frac{4n(n+1)}{2} + n \\ &= \frac{n}{6} \{ (n+1)(2n+1) + 12(n+1) + 6 \} \\ &= \frac{n}{6} \{ (2n^2 + 3n + 1) + (12n + 12) + 6 \} \\ &= \frac{1}{6} n(2n^2 + 15n + 19) \end{aligned}$$

Q.9(iii) Find the sum of n terms of the series whose n th term is:

$$n^3 + 3n^2 + 2n + 1$$

SOLUTION:

$$\begin{aligned} \text{Sum} &= \sum (n^3 + 3n^2 + 2n + 1) \\ &= \sum n^3 + 3 \sum n^2 + 2 \sum n + \sum n^0 \\ &= \left(\frac{n(n+1)}{2} \right)^2 + \frac{3n(n+1)(2n+1)}{6} + \frac{2n(n+1)}{2} + n \\ &= \frac{n^2(n+1)^2}{4} + \frac{n(n+1)(2n+1)}{2} + n(n+1) + n \\ &= \frac{n}{4} \{ n(n+1)^2 + 2(n+1)(2n+1) + 4(n+1) + 4 \} \end{aligned}$$

$$= \frac{n}{4} \{n(n^2 + 2n + 1) + 2(2n^2 + 3n + 1) + (4n + 4) + 4\}$$

$$= \frac{n}{4} \{(n^3 + 2n^2 + n) + (4n^2 + 6n + 2) + (4n + 4) + 4\}$$

$$= \frac{n}{4} (n^3 + 6n^2 + 11n + 10)$$

Q.10(i) Find the sum of first forty six natural numbers starting from 10.

SOLUTION:

$$10 + 11 + 12 + \dots + 55$$

$$= \sum 55 - \sum 9$$

$$= \frac{55(56)}{2} - \frac{9(10)}{2}$$

$$= 1540 - 45$$

$$= 1495$$

Q.10(ii) Find the sum of last 12 terms of the series $1 + 2 + 3 + \dots + 39$.

SOLUTION:

$$\sum 39 - \sum 27$$

$$= \frac{39(40)}{2} - \frac{27(28)}{2}$$

$$= 402$$

Q.11 Find the sum of 9^{th} to 21^{st} terms of the series of squares of first n natural numbers.

SOLUTION:

$$= \sum_{n=1}^{21} n^2 - \sum_{n=1}^8 n^2$$

$$= \frac{21(22)(43)}{6} - \frac{8(9)(17)}{6}$$

$$= 3311 - 204$$

$$= 3107$$

Q.12 Find the sum of last 12 terms of the series of cubes of first 30 natural numbers.

SOLUTION:

$$= \sum_{n=1}^{30} n^3 - \sum_{n=1}^{18} n^3$$

$$= \left(\frac{n(n+1)}{2} \right)^2 - \left(\frac{n(n+1)}{2} \right)^2$$

$$= \left(\frac{30(30+1)}{2} \right)^2 - \left(\frac{18(18+1)}{2} \right)^2$$

$$= (15 \times 31)^2 - (9 \times 19)^2$$

$$= (465)^2 - (171)^2$$

$$= 186984$$

EXERCISE 5.2

Sum the following series up to the indicated terms.

Q.1 $1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + 4 \cdot 2^4 + \dots$ to 50 terms.

SOLUTION:

$$S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + 4 \cdot 2^4 + \dots + 50 \cdot 2^{50}$$

Multiply by Common ratio = 2 and then subtract

$$2S_n = 1 \cdot 2 + 2 \cdot 2^2 + 3 \cdot 2^3 + 4 \cdot 2^4 + \dots + 49 \cdot 2^{50} + 50 \cdot 2^{51}$$

$$-S_n = 2 + \{2^2 + 2^3 + 2^4 + 2^5 + \dots + 2^{50}\} - 50 \cdot 2^{51}$$

$$S_n = -2 - \{a = 2^2, r = 2\} + 50 \cdot 2^{51}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$S_n = -2 - \left\{ \frac{4(2^{49} - 1)}{2 - 1} \right\} + 50 \cdot 2^{51}$$

$$S_n = -2 - 2(2^{50} - 2) + 50 \cdot 2^{51}$$

$$S_n = -2 - 2 \cdot 2^{50} + 4 + 100 \cdot 2^{50}$$

$$S_n = -2 + 4 - 2 \cdot 2^{50} + 100 \cdot 2^{50}$$

$$S_n = 2 + 98 \cdot 2^{50}$$

Q.2 $1 + 2 \cdot 5 + 3 \cdot 5^2 + 4 \cdot 5^3 + \dots$ to 30 terms.

SOLUTION:

$$S_n = 1 + 2 \cdot 5 + 3 \cdot 5^2 + 4 \cdot 5^3 + \dots \text{ to 30 terms.}$$

$$S_n = 1 + 2 \cdot 5 + 3 \cdot 5^2 + 4 \cdot 5^3 + \dots + 30 \cdot 5^{29}$$

Multiply by Common ratio = 5 and then subtract

$$\pm 5S_n = \pm 1 \cdot 5 \pm 2 \cdot 5^2 \pm 3 \cdot 5^3 \pm \dots \pm 29 \cdot 5^{29} \pm 30 \cdot 5^{30}$$

$$-4S_n = 1 + \{5 + 5^2 + 5^3 + \dots + 5^{29}\} - 30 \cdot 5^{30}$$

$$-4S_n = 1 + \{\text{G. P. with } a = 5, r = 5, n = 29\} - 30 \cdot 5^{30}$$

$$S_n = \frac{a(r^n - 1)}{r - 1}$$

$$-4S_n = 1 + \left\{ \frac{5(5^{29} - 1)}{5 - 1} \right\} - 30 \cdot 5^{30}$$

$$-4S_n = 1 + \left\{ \frac{5^{30} - 5}{4} \right\} - 30 \cdot 5^{30}$$

$$-4S_n = \frac{4 + 5^{30} - 5 - 120 \cdot 5^{30}}{4}$$

$$S_n = -\frac{1}{4(4)} \{-119 \cdot 5^{30} - 1\}$$

$$S_n = \frac{1}{16} \{119 \cdot 5^{30} + 1\}$$

Q.3 $1 + \frac{3}{2} + \frac{5}{2^2} + \frac{7}{2^3} + \dots$ to 25 terms.

SOLUTION:

$$S_n = 1 + \frac{3}{2} + \frac{5}{2^2} + \frac{7}{2^3} + \dots \text{ to 25 terms.}$$

$$S_n = 1 + \frac{3}{2} + \frac{5}{2^2} + \frac{7}{2^3} + \dots + \frac{49}{2^{24}}$$

Multiply by Common ratio $= \frac{1}{2}$ and then subtract

$$\frac{1}{2} S_n = \frac{1}{2} + \frac{3}{2^2} + \frac{5}{2^3} + \frac{7}{2^4} + \dots + \frac{47}{2^{24}} + \frac{49}{2^{25}}$$

$$\frac{1}{2} S_n = 1 + \left\{ 1 + \frac{1}{2} + \frac{1}{2^2} + \dots + \frac{1}{2^{23}} \right\} - \frac{49}{2^{25}}$$

$$\frac{1}{2} S_n = 1 + \left\{ \text{G. P. with } a = 1, r = \frac{1}{2}, n = 24 \right\} - \frac{49}{2^{25}}$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$\frac{1}{2} S_n = 1 + \left\{ \frac{\frac{1}{2} \left(1 - \left(\frac{1}{2} \right)^{24} \right)}{1 - \frac{1}{2}} \right\} - \frac{49}{2^{25}}$$

$$\frac{1}{2} S_n = 1 + \left\{ \frac{\frac{1}{2} \left(1 - \frac{1}{2^{24}} \right)}{\frac{1}{2}} \right\} - \frac{49}{2^{25}}$$

$$\frac{1}{2} S_n = 1 + \left(1 - \frac{1}{2^{24}} \right) - \frac{49}{2^{25}}$$

$$S_n = 2 + \left(2 - \frac{1}{2^{23}} \right) - \frac{49}{2^{24}}$$

$$S_n = 4 - \frac{1}{2^{23}} - \frac{49}{2^{24}}$$

$$S_n = 4 - \frac{2}{2^{24}} - \frac{49}{2^{24}}$$

$$S_n = 4 - \frac{51}{2^{24}}$$

$$S_n = 6 - \frac{53}{2^{24}}$$

Q.4 $1 + \frac{8}{5} + \frac{15}{5^2} + \frac{22}{5^3} + \dots$ to n terms.

SOLUTION:

$1 + \frac{8}{5} + \frac{15}{5^2} + \frac{22}{5^3} + \dots$ to n terms

$$1, 8, 15, 22, \dots \rightarrow (\text{A. P. with } a = 1, d = 7)$$

$$1, \frac{1}{5}, \frac{1}{5^2}, \frac{1}{5^3}, \dots \rightarrow (\text{G. P. with } a = 1, r = \frac{1}{5})$$

$$a_n = 1 + 7(n - 1) = 7n - 6$$

$$a_{n-1} = 1 + 7(n - 1) = 7(n - 1) - 6 = 7n - 13$$

$$S_n = 1 + \frac{8}{5} + \frac{15}{5^2} + \frac{22}{5^3} + \dots + \frac{7n-6}{5^{n-1}}$$

Multiply by Common ratio $= \frac{1}{5}$ and then subtract

$$\pm \frac{1}{5} S_n = \pm \frac{1}{5} \pm \frac{8}{5^2} \pm \frac{15}{5^3} \pm \frac{22}{5^4} \pm \dots \pm \frac{7n-13}{5^{n-1}} \pm \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + \left\{ \frac{7}{5} + \frac{7}{5^2} + \frac{7}{5^3} + \dots + \frac{7}{5^{n-1}} \right\} - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + 7 \left\{ \frac{1}{5} + \frac{1}{5^2} + \frac{1}{5^3} + \dots + \frac{1}{5^{n-1}} \right\} - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + 7 \left\{ \text{G. P. with } a = \frac{1}{5}, r = \frac{1}{5}, n = n - 1 \right\} - \frac{7n-6}{5^n}$$

$$S_n = \frac{a(1-r^n)}{1-r}$$

$$\frac{4}{5} S_n = 1 + 7 \left\{ \frac{\frac{1}{5} \left(1 - \left(\frac{1}{5} \right)^{n-1} \right)}{1 - \frac{1}{5}} \right\} - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + 7 \left\{ \frac{\frac{1}{5} \left(1 - \frac{1}{5^{n-1}} \right)}{\frac{4}{5}} \right\} - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + \frac{7}{4} \left(\frac{5^{n-1} - 1}{5^{n-1}} \right) - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + \frac{7}{4} \left(\frac{5^{n-1} - 1}{5^{n-1}} \right) - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + \frac{7}{4} \left(\frac{5^{n-1}}{5^{n-1}} - \frac{1}{5^{n-1}} \right) - \frac{7n-6}{5^n}$$

$$\frac{4}{5} S_n = 1 + \frac{7}{4} \left(1 - \frac{1}{5^{n-1}} \right) - \frac{7n-6}{5^n}$$

$$\frac{4}{5}S_n = 1 + \frac{7}{4} - \frac{7}{4}\left(\frac{1}{5^{n-1}}\right) - \frac{7n-6}{5^n}$$

$$\frac{4}{5}S_n = \frac{11}{4} - \frac{7}{4}\left(\frac{1}{5^{n-1}}\right) - \frac{7n-6}{5^n}$$

$$S_n = \left(\frac{5}{4}\right)\left[\frac{11}{4} - \frac{7}{4}\left(\frac{1}{5^{n-1}}\right) - \frac{7n-6}{5^n}\right]$$

Find the sum to infinity of the following series:

Q.5 $1 + \frac{4}{3} + \frac{9}{3^2} + \frac{16}{3^3} + \frac{25}{3^4} + \dots$

SOLUTION:

$$1 + \frac{4}{3} + \frac{9}{3^2} + \frac{16}{3^3} + \frac{25}{3^4} + \dots$$

$$1, \frac{1}{3}, \frac{1}{3^2}, \frac{1}{3^3}, \dots \rightarrow \left(\text{G. P. with } a = 1, r = \frac{1}{3}\right)$$

$$S_n = 1 + \frac{4}{3} + \frac{9}{3^2} + \frac{16}{3^3} + \frac{25}{3^4} + \dots$$

Multiply by Common ratio $= \frac{1}{3}$ and then subtract

$$\pm \frac{1}{3}S_n = \pm \frac{1}{3} \pm \frac{4}{3^2} \pm \frac{9}{3^3} \pm \frac{16}{3^4} + \dots$$

$$\frac{2}{3}S_n = 1 + \left\{\frac{3}{3} + \frac{5}{3^2} + \frac{7}{3^3} + \frac{9}{3^4} + \dots\right\} \rightarrow (1)$$

$$b_n = \frac{3}{3} + \frac{5}{3^2} + \frac{7}{3^3} + \frac{9}{3^4} + \dots$$

Multiply by Common ratio $= \frac{1}{3}$ and then subtract

$$\frac{1}{3}b_n = \frac{3}{3^2} + \frac{5}{3^3} + \frac{7}{3^4} + \frac{9}{3^5} + \dots$$

$$\frac{2}{3}b_n = 1 + \frac{2}{3^2} + \frac{2}{3^3} + \frac{2}{3^4} + \dots$$

$$\frac{2}{3}b_n = 1 + 2\left\{\frac{1}{3^2} + \frac{1}{3^3} + \frac{1}{3^4} + \dots\right\}$$

$$\frac{2}{3}b_n = 1 + 2\left\{\text{Infinite G. P. with } a = \frac{1}{3^2}, r = \frac{1}{3}\right\}$$

$$\frac{2}{3}b_n = 1 + 2 \left\{ \frac{\frac{1}{9}}{1 - \frac{1}{3}} \right\}$$

$$\frac{2}{3}b_n = 1 + 2 \left\{ \frac{\frac{1}{9}}{\frac{2}{3}} \right\}$$

$$\frac{2}{3}b_n = 1 + \frac{1}{3}$$

$$\frac{2}{3}b_n = \frac{4}{3}$$

$$b_n = \frac{4}{3} \left(\frac{3}{2} \right)$$

$$\boxed{b_n = 2}$$

Put in (1)

$$(1) \Rightarrow \frac{2}{3}S_n = 1 + \{2\}$$

$$\frac{2}{3}S_n = 3$$

$$\boxed{S_n = \frac{9}{2}}$$

Q.6 $1 + 9a + 25a^2 + 49a^3 + \dots$ where $|a| < 1$

SOLUTION:

$$1 + 9a + 25a^2 + 49a^3 + \dots \text{ where } |a| < 1$$

$$1, a, a^2, a^3, \dots \rightarrow (\text{G. P. with } a = 1, r = a)$$

$$S_n = 1 + 9a + 25a^2 + 49a^3 + \dots$$

Multiply by Common ratio = a and then subtract

$$\underline{\pm aS_n = \pm a \pm 9a^2 \pm 25a^3 \pm 49a^4 + \dots}$$

$$S_n - aS_n = 1 + (8a + 16a^2 + 24a^3 + \dots) \rightarrow (1)$$

$$(1 - a)S_n = 1 + (8a + 16a^2 + 24a^3 + \dots) \rightarrow (1)$$

$$t_n = 8a + 16a^2 + 24a^3 + \dots$$

Multiply by Common ratio = a and then subtract

$$\underline{\pm at_n = \pm 8a^2 \pm 16a^3 \pm 24a^4 + \dots}$$

$$t_n - at_n = 8a + 8a^2 + 16a^3 + \dots$$

$$(1 - a)t_n = 8(a + a^2 + a^3 + \dots)$$

$$(1 - a)t_n = 8\{\text{Infinite G. P. with } a = a, r = a\}$$

$$(1 - a)t_n = 8\left(\frac{a}{1-a}\right)$$

$$\boxed{t_n = \frac{8a}{(1-a)^2}}$$

Put in (1)

$$(1) \Rightarrow (1 - a)S_n = 1 + \frac{8a}{(1-a)^2}$$

$$(1 - a)S_n = \frac{(1-a)^2 + 8a}{(1-a)^2}$$

$$S_n = \frac{(1-a)^2 + 8a}{(1-a)^3}$$

$$S_n = \frac{1 - 2a + a^2 + 8a}{(1-a)^3}$$

$$S_n = \frac{1 + 6a + a^2}{(1-a)^3}$$

$$S_n = \frac{1 - a + 7a - 7a^2 + 8a^2}{(1-a)^3}$$

$$\boxed{S_n = \frac{1+7a}{(1-a)^3} + \frac{8a^2}{(1-a)^3}}$$

Q.7 $1 + 4b + 7b^2 + 10b^3 + \dots$ where $|b| < 1$

SOLUTION:

$$1 + 4b + 7b^2 + 10b^3 + \dots \text{ where } |b| < 1$$

$$1, 4, 7, 10, \dots \rightarrow (\text{A. P. with } a = 1, r = 3)$$

$$1, b, b^2, b^3, \dots \rightarrow (\text{G. P. with } a = 1, r = b)$$

$$S_n = 1 + 4b + 7b^2 + 10b^3 + \dots$$

Multiply by Common ratio = b and then subtract,

$$\pm bS_n = \pm b \pm 4b^2 \pm 7b^3 \pm 10b^4 + \dots$$

$$S_n - bS_n = 1 + (3b + 3b^2 + 3b^3 + \dots)$$

$$(1 - b)S_n = 1 + 3(b + b^2 + b^3 + \dots)$$

$$(1 - b)S_n = 1 + 3(\text{Infinite G. P. with } a = b, r = b)$$

$$(1 - b)S_n = 1 + 3\left(\frac{b}{1 - b}\right)$$

$$(1 - b)S_n = \frac{1 - b + 3b}{1 - b}$$

$$S_n = \frac{1 + 2b}{(1 - b)^2}$$

$$S_n = \frac{1 - b + 3b}{(1 - b)^2}$$

$$S_n = \frac{1 - b + 3b}{(1 - b)^2}$$

$$S_n = \frac{1}{1 - b} + \frac{3b}{(1 - b)^2}$$

Q.8 $1 + \frac{4}{4} + \frac{7}{4^2} + \frac{10}{4^3} + \dots$

SOLUTION:

$$1 + \frac{4}{4} + \frac{7}{4^2} + \frac{10}{4^3} + \dots$$

$$1, 4, 7, 10, \dots \rightarrow (\text{G. P. with } a = 1, r = 3)$$

$$1, \frac{1}{4}, \frac{1}{4^2}, \frac{1}{4^3}, \dots \rightarrow \left(\text{G. P. with } a = 1, r = \frac{1}{4} \right)$$

$$S_n = 1 + \frac{4}{4} + \frac{7}{4^2} + \frac{10}{4^3} + \dots$$

Multiply by Common ratio $= \frac{1}{4}$ and then subtract

$$\pm \frac{1}{4} S_n = \pm \frac{1}{4} \pm \frac{4}{4^2} \pm \frac{7}{4^3} \pm \frac{10}{4^4} + \dots$$

$$\frac{3}{4} S_n = 1 + \frac{3}{4} + \frac{3}{4^2} + \frac{3}{4^3} + \dots$$

$$\frac{3}{4} S_n = 1 + 3 \left(\frac{1}{4} + \frac{1}{4^2} + \frac{1}{4^3} + \dots \right)$$

$$\frac{3}{4} S_n = 1 + 3 \left\{ \text{Infinite G. P. with } a = \frac{1}{4}, r = \frac{1}{4} \right\}$$

$$\frac{3}{4} S_n = 1 + 3 \left(\frac{\frac{1}{4}}{1 - \frac{1}{4}} \right)$$

$$\frac{3}{4} S_n = 1 + 3 \left(\frac{\frac{1}{4}}{\frac{3}{4}} \right)$$

$$\frac{3}{4} S_n = 1 + 1$$

$$\frac{3}{4} S_n = 2$$

$$S_n = 2 \left(\frac{4}{3} \right)$$

$$\boxed{S_n = \frac{8}{3}}$$

Q.9 $1 + \frac{5}{3} + \frac{12}{3^2} + \frac{22}{3^3} + \frac{35}{3^4} + \dots$

SOLUTION:

$$1 + \frac{5}{3} + \frac{12}{3^2} + \frac{22}{3^3} + \frac{35}{3^4} + \dots$$

$$1, \frac{1}{3}, \frac{1}{3^2}, \frac{1}{3^3}, \dots \rightarrow \left(\text{G. P. with } a = 1, r = \frac{1}{3} \right)$$

$$S_n = 1 + \frac{5}{3} + \frac{12}{3^2} + \frac{22}{3^3} + \frac{35}{3^4} + \dots$$

Multiply by Common ratio $= \frac{1}{3}$ and then subtract

$$\pm \frac{1}{3} S_n = \pm \frac{1}{3} \pm \frac{5}{3^2} \pm \frac{12}{3^3} \pm \frac{22}{3^4} \pm \dots$$

$$\frac{2}{3} S_n = 1 + \left(\frac{4}{3} + \frac{7}{3^2} + \frac{10}{3^3} + \frac{13}{3^4} + \dots \right) \rightarrow (1)$$

$$a_n = \frac{4}{3} + \frac{7}{3^2} + \frac{10}{3^3} + \frac{13}{3^4} + \dots$$

Multiply by Common ratio $= \frac{1}{3}$ and then subtract

$$\pm \frac{1}{3} a_n = \pm \frac{4}{3^2} \pm \frac{7}{3^3} \pm \frac{10}{3^4} \pm \frac{13}{3^5} + \dots$$

$$\frac{2}{3} a_n = \frac{4}{3} + \frac{3}{3^2} + \frac{3}{3^3} + \frac{3}{3^4} + \dots$$

$$\frac{2}{3} a_n = \frac{4}{3} + \left(\frac{1}{3} + \frac{1}{3^2} + \frac{1}{3^3} + \dots \right)$$

$$\frac{2}{3} a_n = \frac{4}{3} + \left\{ \text{Infinite G. P. with } a = \frac{1}{3}, r = \frac{1}{3} \right\}$$

$$\frac{2}{3} a_n = \frac{4}{3} + \frac{\frac{1}{3}}{1 - \frac{1}{3}}$$

$$\frac{2}{3} a_n = \frac{4}{3} + \frac{\frac{1}{3}}{\frac{2}{3}}$$

$$\frac{2}{3} a_n = \frac{4}{3} + \frac{1}{2}$$

$$\frac{2}{3} a_n = \frac{11}{6}$$

$$a_n = \frac{11}{6} \left(\frac{3}{2} \right)$$

$$a_n = \frac{11}{4}$$

Put in (1)

$$(1) \Rightarrow \frac{2}{3} S_n = 1 + \frac{11}{4}$$

$$\frac{2}{3} S_n = \frac{15}{4}$$

$$S_n = \frac{15}{4} \left(\frac{3}{2} \right)$$

$$S_n = \frac{45}{8}$$

Q.10 $1 + 5a + 9a^2 + 13a^3 + \dots$ where $|a| < 1$

SOLUTION:

$1 + 5a + 9a^2 + 13a^3 + \dots$ where $|a| < 1$

$1, 5, 9, 13, \dots \rightarrow$ (A. P. with $a = 1, r = 4$).

$1, a, a^2, a^3, \dots \rightarrow$ (G. P. with $a = 1, r = a$)

$$S_n = 1 + 5a + 9a^2 + 13a^3 + \dots$$

Multiply by Common ratio = a and then subtract

$$\underline{\pm a S_n = \pm a \pm 5a^2 \pm 9a^3 \pm 13a^4 + \dots}$$

$$S_n - a S_n = 1 + 4a + 4a^2 + 4a^3 + \dots$$

$$(1 - a) S_n = 1 + 4(a + a^2 + a^3 + \dots)$$

$$(1 - a) S_n = 1 + 4(\text{Infinite G. P. with } a = a, r = a)$$

$$(1 - a) S_n = 1 + 4 \left(\frac{a}{1 - a} \right)$$

$$(1 - a) S_n = \frac{1 - a + 4a}{1 - a}$$

$$(1 - a) S_n = \frac{1 + 3a}{1 - a}$$

$$S_n = \frac{1 + 3a}{(1 - a)^2}$$

$$S_n = \frac{1 - a + 4a}{(1 - a)^2}$$

$$S_n = \frac{1}{1-a} + \frac{4a}{(1-a)^2}$$

EXERCISE 5.3

Find the sum of the following series:

Q.1 $3 + 6 + 12 + 21 + 33 + \dots$ to n terms

SOLUTION:

$3 + 6 + 12 + 21 + 33 + \dots$ to n terms

$$(6 - 3) = 3$$

$$(12 - 6) = 6$$

$$(21 - 12) = 9$$

$$(33 - 21) = 12$$

Here, differences of consecutive terms are in A.P.

$S_n = 3 + 6 + 12 + 21 + 33 + \dots$ to n terms

$$S_n = 3 + 6 + 12 + 21 + 33 + \dots + a_n$$

$$\pm S_n = \pm 3 \pm 6 \pm 12 \pm 21 \pm 33 + \dots \pm a_{n-1} \pm a_n$$

$$0 = 3 + 3 + 6 + 9 + 12 + \dots - a_n$$

$$a_n = 3 + \{3 + 6 + 9 + 12 + \dots \text{to } (n-1) \text{ terms}\}$$

$$a_n = 3 + \{\text{A.P. with } a = 3, d = 3, \text{no. of terms} = (n-1)\}$$

$$a_n = 3 + \frac{(n-1)}{2} \{2(3) + 3(n-1-1)\}$$

$$a_n = 3 + \frac{(n-1)}{2} \{6 + 3n - 6\}$$

$$a_n = 3 + \frac{(n-1)}{2} \{3n\}$$

$$a_n = \frac{6+3n(n-1)}{2}$$

$$a_n = \frac{6+3n^2-3n}{2}$$

$$a_n = \frac{3n^2-3n+6}{2}$$

$$S_n = \frac{1}{2} \sum (3n^2 - 3n + 6)$$

$$S_n = \frac{1}{2} \left\{ 3 \sum n^2 - 3 \sum n + 6 \sum n^0 \right\}$$

$$S_n = \frac{1}{2} \left\{ \frac{3n(n+1)(2n+1)}{6} - 3 \frac{n(n+1)}{2} + 6n \right\}$$

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$$S_n = \frac{n}{2} \left\{ \frac{(n+1)(2n+1)}{2} - \frac{3(n+1)}{2} + 6 \right\}$$

$$S_n = \frac{n}{2} \left\{ \frac{(2n^2 + 3n + 1) - 3(n+1) + 12}{2} \right\}$$

$$S_n = \frac{n}{2(2)} \{2n^2 + 3n + 1 - 3n - 3 + 12\}$$

$$S_n = \frac{n}{4} (2n^2 + 10)$$

$$S_n = \frac{n}{4} (2)(n^2 + 5)$$

$$S_n = \frac{n}{2} (n^2 + 5)$$

Q.25 $5 + 10 + 17 + 26 + 37 + \dots$ to n terms

SOLUTION:

$5 + 10 + 17 + 26 + 37 + \dots$ to n terms

$$(10 - 5) = 5$$

$$(17 - 10) = 7$$

$$(26 - 17) = 9$$

$$(37 - 26) = 11$$

Here, differences of consecutive terms are in A.P.

$S_n = 5 + 10 + 17 + 26 + 37 + \dots$ to n terms

$$S_n = 5 + 10 + 17 + 26 + 37 + \dots + a_n$$

$$\pm S_n = \pm 5 \pm 10 \pm 17 \pm 26 \pm 37 \pm \dots \pm a_{n-1} \pm a_n$$

$$0 = 5 + 5 + 7 + 9 + 11 + \dots + a_{n-1} - a_n$$

$$a_n = 5 + (5 + 7 + 9 + 11 + \dots + \text{to } (n-1) \text{ terms})$$

$$a_n = 5 + \{\text{A.P. with } a = 5, d = 2, \text{ no. of terms} = (n-1)\}$$

$$a_n = 5 + \frac{(n-1)}{2} \{2(5) + 2(n-1-1)\}$$

$$a_n = 5 + \frac{(n-1)}{2} (2) \{5 + n - 2\}$$

$$a_n = 5 + (n-1)(n+3)$$

$$a_n = 5 + n^2 + 2n - 3$$

$$a_n = n^2 + 2n + 2$$

$$S_n = \sum (n^2 + 2n + 2)$$

$$S_n = \sum n^2 + 2 \sum n + 2 \sum n^0$$

$$S_n = \frac{n(n+1)(2n+1)}{6} + 2 \frac{n(n+1)}{2} + 2n$$

$$S_n = \frac{(n+1)(2n+1)}{6} + n(n+1) + 2n$$

$$S_n = \frac{n(n+1)(2n+1) + 6n(n+1) + 12n}{6}$$

$$S_n = \frac{n}{6} \{ (n+1)(2n+1) + 6(n+1) + 12 \}$$

$$S_n = \frac{n}{6} \{ (2n^2 + 3n + 1) + 6n + 6 + 12 \}$$

$$S_n = \frac{n}{6} (2n^2 + 9n + 19)$$

Q.37 + 14 + 26 + 43 + 65 + ... to n terms

SOLUTION:

7 + 14 + 26 + 43 + 65 + ... to n terms

$$(14 - 7) = 7$$

$$(26 - 14) = 12$$

$$(43 - 26) = 17$$

$$(65 - 43) = 22$$

Here, differences of consecutive terms are in A.P.

$S_n = 7 + 14 + 26 + 43 + 65 + \dots$ to n terms

$S_n = 7 + 14 + 26 + 43 + 65 + \dots + a_n$

$$\pm S_n = \pm 7 \pm 14 \pm 26 \pm 43 \pm 65 \pm \dots \pm a_{n-1} \pm a_n$$

$$0 = 7 + 7 + 12 + 17 + 22 + \dots - a_n$$

$$a_n = 7 + \{7 + 12 + 17 + 22 + \dots \text{ to } (n-1) \text{ terms}\}$$

$$a_n = 7 + \{ \text{A.P. with } a = 7, d = 5, \text{ no. of terms} = (n-1) \}$$

$$a_n = 7 + \frac{(n-1)}{2} \{ 2(7) + 5(n-1-1) \}$$

$$a_n = 7 + \frac{(n-1)}{2} \{ 14 + 5n - 10 \}$$

$$a_n = 7 + \frac{(n-1)}{2} (4 + 5n)$$

$$a_n = \frac{14 + 5n^2 - n - 4}{2}$$

$$a_n = \frac{5n^2 - n + 10}{2}$$

$$S_n = \frac{1}{2} \sum (5n^2 - n + 10)$$

$$S_n = \frac{1}{2} \left\{ 5 \sum n^2 - \sum n + 10 \sum n^0 \right\}$$

$$S_n = \frac{1}{2} \left\{ 5 \frac{n(n+1)(2n+1)}{6} - \frac{n(n+1)}{2} + 10n \right\}$$

$$S_n = \frac{1}{2} \left\{ \frac{5n(n+1)(2n+1) - 3n(n+1) + 60n}{6} \right\}$$

$$S_n = \frac{n}{2} \left\{ \frac{5(n+1)(2n+1) - 3(n+1) + 60}{6} \right\}$$

$$S_n = \frac{n}{12} \{ 5(2n^2 + 3n + 1) - 3n + 57 \}$$

$$S_n = \frac{n}{12} \{ 10n^2 + 15n + 5 - 3n + 57 \}$$

$$S_n = \frac{n}{12} (10n^2 + 12n + 62)$$

$$S_n = \frac{n}{12} (2)(5n^2 + 6n + 31)$$

$$S_n = \frac{n}{6} (5n^2 + 6n + 31)$$

Q.43 $3 + 6 + 15 + 42 + 123 + \dots$ to n terms

SOLUTION:

$3 + 6 + 15 + 42 + 123 + \dots$ to n terms

$$(6 - 3) = 3$$

$$(15 - 6) = 9$$

$$(42 - 15) = 27$$

$$(123 - 42) = 81$$

Here, differences of consecutive terms are in G.P.

$$S_n = 3 + 6 + 15 + 42 + 123 + \dots \text{ to } n \text{ terms}$$

$$S_n = 3 + 6 + 15 + 42 + 123 + \dots + a_n$$

$$\pm S_n = \pm 3 \pm 6 \pm 15 \pm 42 \pm 123 + \dots \pm a_{n-1} \pm a_n$$

$$0 = 3 + 3 + 9 + 27 + 81 + \dots - a_n$$

$$a_n = 3 + \{ 3 + 6 + 9 + 27 + 81 + \dots \text{ to } (n-1) \text{ terms} \}$$

$$a_n = 3 + \left\{ \frac{3(3^{n-1} - 1)}{3 - 1} \right\}$$

$$a_n = 3 + \frac{3^n - 3}{2}$$

$$a_n = \frac{6+3^n-3}{2}$$

$$a_n = \frac{3+3^n}{2}$$

$$S_n = \frac{1}{2} \sum (3 + 3^n)$$

$$S_n = \frac{1}{2} \left\{ 3 \sum n^0 + \sum 3^n \right\}$$

$$S_n = \frac{1}{2} \{ 3n + (3 + 3^2 + 3^3 + \dots + 3^n) \}$$

$$S_n = \frac{1}{2} \left\{ 3n + \frac{3(3^n-1)}{3-1} \right\}$$

$$S_n = \frac{1}{2} \left\{ 3n + \frac{3^{n+1}-3}{2} \right\}$$

$$S_n = \frac{1}{2} \left\{ \frac{6n+3^{n+1}-3}{2} \right\}$$

$$S_n = \frac{1}{4} (3^{n+1} + 6n - 3)$$

$$S_n = \frac{1}{4} (6n + 3(3^n - 1))$$

Q.5 $7 + 14 + 45 + 100 + 179 + \dots$ to n terms

SOLUTION:

$7 + 14 + 45 + 100 + 179 + \dots$ to n terms

$$(14 - 7) = 7$$

$$(45 - 14) = 31$$

$$(100 - 45) = 55$$

$$(179 - 100) = 79$$

Here, differences of consecutive terms are in G.P.

$$S_n = 7 + 14 + 45 + 100 + 179 + \dots \text{ to } n \text{ terms}$$

$$S_n = 7 + 14 + 45 + 100 + 179 + \dots + a_n$$

$$\pm S_n = \pm 7 \pm 14 \pm 45 \pm 100 \pm 179 \pm \dots \pm a_{n-1} \pm a_n$$

$$0 = 7 + 7 + 31 + 55 + 79 + \dots - a_n$$

$$a_n = 7 + \{7 + 31 + 55 + 79 + \dots \text{ to } (n-1) \text{ terms}\}$$

$$a_n = 7 + \{a = 7, d = 24, \text{ no. of terms} = (n-1)\}$$

$$a_n = 7 + \frac{(n-1)}{2} \{2(7) + 24(n-1-1)\}$$

$$a_n = 7 + \frac{(n-1)}{2} \{14 + 24(n-2)\}$$

$$a_n = 7 + \frac{(n-1)}{2} \{14 + 24n - 48\}$$

$$a_n = 7 + \frac{(n-1)}{2} \{24n - 34\}$$

$$a_n = 7 + (n-1)(12n-17)$$

$$a_n = 7 + 12n^2 - 17n - 12n + 17$$

$$a_n = 24 + 12n^2 - 29n$$

$$a_n = 12n^2 - 29n + 24$$

$$S_n = \sum (12n^2 - 29n + 24)$$

$$S_n = \left\{ 12 \sum n^2 - 29 \sum n + 24 \sum n^0 \right\}$$

$$S_n = \left\{ 12 \frac{n(n+1)(2n+1)}{6} - 29 \frac{n(n+1)}{2} + 24n \right\}$$

$$S_n = \left\{ \frac{4n(n+1)(2n+1) - 29n(n+1) + 48n}{2} \right\}$$

$$S_n = \frac{n}{2} \{4(n+1)(2n+1) - 29(n+1) + 48\}$$

$$S_n = \frac{n}{2} \{4(2n^2 + 3n + 1) - 29n - 29 + 48\}$$

$$S_n = \frac{n}{2} \{8n^2 + 12n + 4 - 29n + 19\}$$

$$S_n = \frac{n}{2} (8n^2 - 17n + 23)$$

Q.6 $3 + 6 + 21 + 96 + 471 + \dots$ to n terms

SOLUTION:

$3 + 6 + 21 + 96 + 471 + \dots$ to n terms

$$(6 - 3) = 3$$

$$(21 - 6) = 15$$

$$(96 - 21) = 75$$

$$(471 - 96) = 375$$

Here, differences of consecutive terms are in G.P.

$$S_n = 3 + 6 + 21 + 96 + 471 + \dots \text{ to } n \text{ terms}$$

$$S_n = 3 + 6 + 21 + 96 + 471 + \dots + a_n$$

$$\pm S_n = \pm 3 \pm 6 \pm 21 \pm 96 \pm 471 \pm \dots \pm a_{n-1} \pm a_n$$

$$0 = 3 + 3 + 15 + 75 + 375 + \dots - a_n$$

$$a_n = 3 + \{3 + 15 + 75 + 375 + \dots \text{to } (n) \text{ terms}\}$$

$$a_n = 3 + \{\text{G. P. with } a = 3, r = 5, \text{ no. of terms} = (n)\}$$

$$a_n = 3 + \left\{ \frac{3(5^{n-1}-1)}{5-1} \right\}$$

$$a_n = 3 + \frac{3(5^{n-1}-1)}{4}$$

$$a_n = \frac{12+3(5^{n-1}-1)}{4}$$

$$a_n = \frac{12+(3)5^{n-1}-3}{4}$$

$$a_n = \frac{9+(3)5^{n-1}}{4}$$

$$S_n = \frac{1}{4} \sum (9 + (3)5^{n-1})$$

$$S_n = \frac{1}{4} \left\{ 9 \sum n^0 + \frac{3}{5} \sum 5^n \right\}$$

$$S_n = \frac{1}{4} \left\{ 9n + \frac{3}{5} (5 + 5^2 + \dots \text{to } n \text{ terms}) \right\}$$

$$S_n = \frac{1}{4} \left\{ 9n + \frac{3}{5} (\text{G. P. with } a = 5, r = 5, n = n) \right\}$$

$$S_n = \frac{1}{4} \left\{ 9n + \frac{3}{5} (5) \left(\frac{5^n-1}{5-1} \right) \right\}$$

$$S_n = \frac{1}{4} \left\{ 9n + 3 \left(\frac{5^n-1}{4} \right) \right\}$$

$$S_n = \frac{1}{4} \left\{ \frac{36n+(3)5^n-3}{20} \right\}$$

$$S_n = \left\{ \frac{36n+(3)(5^n-1)}{16} \right\}$$

EXERCISE 5.4

Find the sum of the following series:

Q.1 $\frac{1}{1 \cdot 3} + \frac{1}{3 \cdot 5} + \frac{1}{5 \cdot 7} + \dots$ to n terms.

SOLUTION:

1, 3, 5, ... (A. P. with $a = 1, d = 2$)

3, 5, 7, ... (A. P. with $a = 3, d = 2$)

$$a_n = a + (n-1)d$$

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$$a_n = 1 + 2(n-1) = 1 + 2n - 2 = 2n - 1$$

$$a_n = 3 + 2(n-1) = 3 + 2n - 2 = 2n + 1$$

$$nth \text{ term} = \frac{1}{(2n-1)(2n+1)}$$

$$\frac{1}{(2n-1)(2n+1)} = \frac{(2n+1)-(2n-1)}{2(2n-1)(2n+1)}$$

$$= \frac{1}{2} \left\{ \frac{(2n+1)}{(2n-1)(2n+1)} - \frac{(2n-1)}{(2n-1)(2n+1)} \right\}$$

$$= \frac{1}{2} \left\{ \frac{1}{2n-1} - \frac{1}{2n+1} \right\}$$

$$\text{Sum} = \sum \frac{1}{(2n-1)(2n+1)}$$

$$\text{Sum} = \frac{1}{2} \sum \left\{ \frac{1}{2n-1} - \frac{1}{2n+1} \right\}$$

$$f(r) = \frac{1}{2r-1}$$

$$\text{Sum} = \frac{1}{2} \{f(1) - f(n+1)\}$$

$$= \frac{1}{2} \{f(1) - f(n+1)\}$$

$$= \frac{1}{2} \left\{ \frac{1}{2(1)-1} - \frac{1}{2(n+1)-1} \right\}$$

$$= \frac{1}{2} \left\{ \frac{1}{1} - \frac{1}{2n+2-1} \right\}$$

$$= \frac{1}{2} \left\{ 1 - \frac{1}{2n+1} \right\}$$

$$= \frac{1}{2} \left\{ \frac{2n+1-1}{2n+1} \right\}$$

$$= \frac{1}{2} \left\{ \frac{2n}{2n+1} \right\}$$

$$= \frac{n}{2n+1}$$

Q.2 $\frac{1}{1 \cdot 5} + \frac{1}{5 \cdot 9} + \frac{1}{9 \cdot 13} + \dots$ to n terms.

SOLUTION:

$$\frac{1}{1 \cdot 5} + \frac{1}{5 \cdot 9} + \frac{1}{9 \cdot 13} + \dots \text{ to } n \text{ terms}$$

$$1, 5, 9, \dots \text{ (A. P. with } a = 1, d = 4)$$

$$5, 9, 13, \dots \text{ (A. P. with } a = 5, d = 4)$$

$$a_n = a + (n-1)d$$

$$a_n = 1 + 4(n - 1) = 1 + 4n - 4 = 4n - 3$$

$$a_n = 5 + 4(n - 1) = 5 + 4n - 4 = 4n + 1$$

$$nth \text{ term} = \frac{1}{(4n-3)(4n+1)}$$

$$= \frac{(4n+1) - (4n-3)}{4(4n-3)(4n+1)}$$

$$= \frac{1}{4} \left\{ \frac{(4n+1)}{(4n-3)(4n+1)} - \frac{(4n-3)}{(4n-3)(4n+1)} \right\}$$

$$= \frac{1}{4} \left\{ \frac{1}{4n-3} - \frac{1}{4n+1} \right\}$$

$$\text{Sum} = \sum \frac{1}{(4n-3)(4n+1)}$$

$$\text{Sum} = \frac{1}{4} \sum \left\{ \frac{1}{4n-3} - \frac{1}{4n+1} \right\}$$

$$f(r) = \frac{1}{4r-3}$$

$$\text{Sum} = \frac{1}{4} \{f(1) - f(n+1)\}$$

$$= \frac{1}{4} \left\{ \frac{1}{4(1)-3} - \frac{1}{4(n+1)-3} \right\}$$

$$= \frac{1}{4} \left\{ \frac{1}{1} - \frac{1}{4n+4-3} \right\}$$

$$= \frac{1}{4} \left\{ \frac{1}{1} - \frac{1}{4n+1} \right\}$$

$$= \frac{1}{4} \left\{ \frac{4n+1-1}{4n+1} \right\}$$

$$= \frac{1}{4} \left\{ \frac{4n}{4n+1} \right\}$$

$$= \frac{n}{4n+1}$$

Q.3 $\frac{1}{1 \cdot 5 \cdot 9} + \frac{1}{5 \cdot 9 \cdot 13} + \frac{1}{9 \cdot 13 \cdot 17} + \dots$ to infinity.

SOLUTION:

$$\frac{1}{1 \cdot 5 \cdot 9} + \frac{1}{5 \cdot 9 \cdot 13} + \frac{1}{9 \cdot 13 \cdot 17} + \dots \text{ to infinity}$$

$$1, 5, 9, \dots \text{ (A. P. with } a = 1, d = 4)$$

$$5, 9, 13, \dots \text{ (A. P. with } a = 5, d = 4)$$

$$9, 13, 17, \dots \text{ (A. P. with } a = 9, d = 4)$$

$$a_n = a + (n - 1)d$$

$$a_n = 1 + 4(n-1) = 1 + 4n - 4 = 4n - 3$$

$$a_n = 5 + 4(n-1) = 5 + 4n - 4 = 4n + 1$$

$$a_n = 9 + 4(n-1) = 9 + 4n - 4 = 4n + 5$$

$$nth \text{ term} = \frac{1}{(4n-3)(4n+1)(4n+5)}$$

$$\frac{1}{(4n-3)(4n+1)(4n+5)} = \frac{A}{4n+1} + \frac{B}{4n-3} + \frac{C}{4n+5}$$

$$\frac{1}{(4n-3)(4n+1)(4n+5)} = \frac{-\frac{1}{16}}{4n+1} + \frac{\frac{1}{32}}{4n-3} + \frac{\frac{1}{32}}{4n+5}$$

$$\text{Sum} = \sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)(4n+5)}$$

$$\begin{aligned} \text{Sum} &= \sum_{n=1}^{\infty} \left\{ \frac{1}{32(4n-3)} + \frac{1}{32(4n+5)} - \frac{1}{16(4n+1)} \right\} \\ &= \left(\frac{1}{32} + \frac{1}{160} + \frac{1}{288} + \frac{1}{416} + \dots \right) + \left(\frac{1}{288} + \frac{1}{416} + \frac{1}{544} + \frac{1}{672} + \dots \right) \\ &\quad - \left(\frac{1}{80} + \frac{1}{144} + \frac{1}{208} + \frac{1}{272} + \dots \right) \\ &= \frac{1}{40} \end{aligned}$$

Q.4 $\frac{1}{1 \cdot 4 \cdot 7} + \frac{1}{4 \cdot 7 \cdot 10} + \frac{1}{7 \cdot 10 \cdot 13} + \dots$ to infinity.

SOLUTION:

$$\frac{1}{1 \cdot 4 \cdot 7} + \frac{1}{4 \cdot 7 \cdot 10} + \frac{1}{7 \cdot 10 \cdot 13} + \dots \text{ to infinity}$$

$$1, 4, 7, \dots \text{ (A. P. with } a = 1, d = 3)$$

$$4, 7, 10, \dots \text{ (A. P. with } a = 4, d = 3)$$

$$7, 10, 13, \dots \text{ (A. P. with } a = 7, d = 3)$$

$$a_n = a + (n-1)d$$

$$a_n = 1 + 3(n-1) = 1 + 3n - 3 = 3n - 2$$

$$a_n = 4 + 3(n-1) = 4 + 3n - 3 = 3n + 1$$

$$a_n = 7 + 3(n-1) = 7 + 3n - 3 = 3n + 4$$

$$nth \text{ term} = \frac{1}{(3n-2)(3n+1)(3n+4)}$$

$$\frac{1}{(3n-2)(3n+1)(3n+4)} = \frac{A}{3n+1} + \frac{A}{3n-2} + \frac{C}{3n+4}$$

$$\frac{1}{(4n-3)(4n+1)(4n+5)} = \frac{-\frac{1}{9}}{3n+1} + \frac{\frac{1}{18}}{3n-2} + \frac{\frac{1}{18}}{3n+4}$$

$$\text{Sum} = \sum_{n=1}^{\infty} \frac{1}{(4n-3)(4n+1)(4n+5)}$$

$$\text{Sum} = \sum_{n=1}^{\infty} \left\{ \frac{1}{18(3n-2)} + \frac{1}{18(3n+4)} - \frac{1}{9(3n+1)} \right\}$$

$$= \left(\frac{1}{18} + \frac{1}{72} + \frac{1}{126} + \frac{1}{180} + \dots \right) + \left(\frac{1}{126} + \frac{1}{180} + \frac{1}{234} + \frac{1}{288} + \dots \right)$$

$$- \left(\frac{1}{36} + \frac{1}{63} + \frac{1}{90} + \frac{1}{117} + \dots \right)$$

$$= \frac{1}{24}$$

Find the sum of the series:

Q.5 $\sum_{k=1}^{\infty} \frac{1}{4k^2-1}$

SOLUTION:

$$\sum_{k=1}^{\infty} \frac{1}{4k^2-1} = \sum_{k=1}^{\infty} \frac{1}{(2k)^2-(1)^2}$$

$$= \sum_{k=1}^{\infty} \frac{1}{(2k-1)(2k+1)}$$

$$\frac{1}{(2k-1)(2k+1)} = \frac{(2k+1)-(2k-1)}{2(2k-1)(2k+1)}$$

$$= \frac{1}{2} \left\{ \frac{(2k+1)}{(2k-1)(2k+1)} - \frac{(2k-1)}{(2k-1)(2k+1)} \right\}$$

$$= \frac{1}{2} \left\{ \frac{1}{2k-1} - \frac{1}{2k+1} \right\}$$

$$\begin{aligned}\sum_{k=1}^{\infty} \frac{1}{4k^2 - 1} &= \frac{1}{2} \sum_{k=1}^{\infty} \left\{ \frac{1}{2k-1} - \frac{1}{2k+1} \right\} \\ &= \frac{1}{2} \left\{ \left(1 + \frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots \right) - \left(\frac{1}{3} + \frac{1}{5} + \frac{1}{7} + \dots \right) \right\} \\ &= \frac{1}{2} (1) \\ &= \frac{1}{2}\end{aligned}$$

Find the sum of the series:

Q. 6 $\sum_{k=1}^n \frac{1}{9k^2 + 3k - 2}$

SOLUTION:

$$\sum_{k=1}^n \frac{1}{9k^2 + 3k - 2}$$

$$\begin{aligned}9k^2 + 3k - 2 &= 9k^2 + 6k - 3k - 2 = 3k(3k + 2) - (3k + 2) \\ &= (3k + 2)(3k - 1)\end{aligned}$$

$$= \sum_{k=1}^n \frac{1}{(3k + 2)(3k - 1)}$$

$$= \sum_{k=1}^n \frac{(3k + 2) - (3k - 1)}{3(3k + 2)(3k - 1)}$$

$$= \frac{1}{3} \sum_{k=1}^n \frac{(3k + 2) - (3k - 1)}{(3k + 2)(3k - 1)}$$

$$= \frac{1}{3} \sum_{k=1}^n \left\{ \frac{(3k + 2) - (3k - 1)}{(3k + 2)(3k - 1)} \right\}$$

$$= \frac{1}{3} \sum_{k=1}^n \left\{ \frac{(3k+2)}{(3k+2)(3k-1)} - \frac{(3k-1)}{(3k+2)(3k-1)} \right\}$$

$$= \frac{1}{3} \sum_{k=1}^n \left\{ \frac{1}{3k-1} - \frac{1}{3k+2} \right\}$$

$$f(r) = \frac{1}{3r-1}$$

$$= \frac{1}{3} \left\{ \frac{1}{3(1)-1} - \frac{1}{3(n+1)-1} \right\}$$

$$= \frac{1}{3} \left\{ \frac{1}{2} - \frac{1}{3n+3-1} \right\}$$

$$= \frac{1}{3} \left\{ \frac{1}{2} - \frac{1}{3n+2} \right\}$$

$$= \frac{1}{3} \left\{ \frac{3n+2-2}{3n+2} \right\}$$

$$= \frac{1}{3} \left\{ \frac{3n}{3n+2} \right\}$$

$$= \frac{n}{3n+2}$$

Find the sum of the series:

$$Q.7 \sum_{k=2}^{\infty} \frac{1}{k^2 - 1}$$

SOLUTION:

$$\sum_{k=2}^{\infty} \frac{1}{k^2 - 1} = \sum_{k=2}^{\infty} \frac{1}{(k)^2 - (1)^2}$$

$$= \sum_{k=2}^{\infty} \frac{1}{(k-1)(k+1)}$$

$$= \sum_{k=2}^{\infty} \frac{(k+1) - (k-1)}{2(k-1)(k+1)}$$

$$= \frac{1}{2} \sum_{k=2}^{\infty} \left\{ \frac{(k+1)}{(k-1)(k+1)} - \frac{(k-1)}{(k-1)(k+1)} \right\}$$

$$= \frac{1}{2} \sum_{k=2}^{\infty} \left\{ \frac{1}{k-1} - \frac{1}{k+1} \right\}$$

$$= \frac{1}{2} \left\{ \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots \right) - \left(\frac{1}{3} + \frac{1}{4} + \frac{1}{5} + \dots \right) \right\}$$

$$= \frac{1}{2} \left\{ +1 + \frac{1}{2} \right\}$$

$$= \frac{1}{2} \left\{ \frac{3}{2} \right\}$$

$$= \frac{3}{4}$$

Find the sum of the series:

Q. 8 $\sum_{k=2}^n \frac{1}{k^2 - k}$

SOLUTION:

$$\sum_{k=2}^n \frac{1}{k^2 - k} = \sum_{k=2}^n \frac{1}{k(k-1)}$$

$$= \sum_{k=2}^n \frac{k - (k-1)}{k(k-1)}$$

$$= \sum_{k=2}^n \left\{ \frac{k}{k(k-1)} - \frac{(k-1)}{k(k-1)} \right\}$$

$$= \sum_{k=2}^n \left\{ \frac{1}{k-1} - \frac{1}{k} \right\}$$

$$= \left(1 + \frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n-1} \right) - \left(\frac{1}{2} + \frac{1}{3} + \frac{1}{4} + \dots + \frac{1}{n} \right)$$

$$= 1 - \frac{1}{n}$$

$$= \frac{n-1}{n}$$

REVIEW EXERCISE 5

1. Select correct option.

(i) If $a_n = 5 - 3n + 2n^2$, then $a_{2n} =$ _____

(a) $5 - 6n + 2n^2$

(b) $5 - 6n + 4n^2$

(c) $5 + 6n + 4n^2$

(d) $5 - 6n + 8n^2$

(ii) If $a_{n-2} = 3n - 11$, then $a_n =$ _____

(a) $3n + 5$

(b) $3n - 5$

(c) $3n - 9$

(d) $3n - 13$

(iii) $\sum_{n=1}^{50} n =$ _____

(a) 1274

(b) 1275

(c) 1280

(d) 1285

(iv) $\frac{1}{n(n+1)} =$ _____

(a) $\frac{1}{n} + \frac{1}{n+1}$

(b) $\frac{1}{n} - \frac{1}{n+1}$

(c) $\frac{2}{n} + \frac{1}{n+1}$

(d) $\frac{1}{n} - \frac{3}{n+1}$

(v) the n th term of $\frac{1}{2}, \frac{1}{5}, \frac{1}{8}, \dots$ is:

(a) $\frac{1}{3n-1}$

(b) $3n - 1$

(c) $2n + 1$

(d) $\frac{1}{3n+1}$

(vi) $\sum n$ is equal to

(a) $\frac{n(n+1)}{2}$

(b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{n^2(n+1)^2}{2}$

(d) n^2

(vii) $\sum n^2$ is equal to

(a) $\frac{n(n+1)}{2}$

(b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{n^2(n+1)^2}{2}$

(d) n^2

(viii) $\sum n^3$ is equal to

(a) $\frac{n(n+1)}{2}$

(b) $\frac{n(n+1)(2n+1)}{6}$

(c) $\frac{n^2(n+1)^2}{2}$

(d) n^2

(ix) If $S_n = (n+1)^2$, then S_{2n} is equal to:

(a) $2n+1$

(b) $4n^2 + 4n + 1$

(c) $(2n-1)^2$

(d) cannot be determined

(x) $\sum_{n=3}^{20} n^0 = \underline{\hspace{2cm}}$

(a) 1

(b) 19

(c) 20

(d) 18

Q.2 Sum the following series up to n terms

$$2 \times 1^2 + 4 \times 2^2 + 6 \times 3^2 + \dots$$

SOLUTION:

$$2 \times 1^2 + 4 \times 2^2 + 6 \times 3^2 + \dots$$

2, 4, 6, ... (A. P. with $a = 2, d = 2$)

$$a_n = a + (n - 1)d$$

$$a_n = 2 + 2(n - 1) = 2 + 2n - 2 = 2n$$

$$nth \text{ term} = (2n)(n^2) = 2n^3$$

$$\text{Sum} = \sum_{n=1}^n (2n^3)$$

$$= 2 \sum_{n=1}^n n^3$$

$$= 2 \left\{ \frac{n(n+1)}{2} \right\}^2$$

$$= \frac{2n^2(n+1)^2}{4}$$

$$= \frac{n^2(n+1)^2}{2}$$

Q.3 Sum the series $\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \dots$ to infinity

SOLUTION:

$$\frac{1}{1 \cdot 4} + \frac{1}{4 \cdot 7} + \frac{1}{7 \cdot 10} + \dots \text{ to infinity}$$

1, 4, 7, ... (A. P. with $a = 1, d = 3$)

4, 7, 10, ... (A. P. with $a = 4, d = 3$)

7, 10, 13, ... (A. P. with $a = 7, d = 3$)

$$a_n = a + (n - 1)d$$

$$a_n = 1 + 3(n - 1) = 1 + 3n - 3 = 3n - 2$$

$$a_n = 4 + 3(n - 1) = 4 + 3n - 3 = 3n + 1$$

$$nth \text{ term} = \frac{1}{(3n-2)(3n+1)}$$

$$\text{Sum} = \sum_{n=1}^{\infty} \frac{1}{(3n-2)(3n+1)}$$

$$= \sum_{n=1}^{\infty} \frac{(3n+1) - (3n-2)}{3(3n-2)(3n+1)}$$

$$= \frac{1}{3} \sum_{n=1}^{\infty} \left\{ \frac{(3n+1)}{(3n-2)(3n+1)} - \frac{(3n-2)}{(3n-2)(3n+1)} \right\}$$

$$= \frac{1}{3} \sum_{n=1}^{\infty} \left\{ \frac{1}{3n-2} - \frac{1}{3n+1} \right\}$$

$$= \frac{1}{3} \left\{ \left(1 + \frac{1}{4} + \frac{1}{7} + \frac{1}{10} + \dots \right) - \left(\frac{1}{4} + \frac{1}{7} + \frac{1}{10} + \dots \right) \right\}$$

$$= \frac{1}{3} (1)$$

$$= \frac{1}{3}$$